Selected Social Problem

Topic: Cybersecurity

Research Title

Title: Cybersecurity in Mobile Banking: Analyzing Phishing Attacks and Security Challenges in Bangladesh.

Introduction

Nowadays, mobile financial services like bKash, Nagad, and Rocket are very popular in Bangladesh. Almost everyone uses them for daily transactions. While this digital change has made our life very easy, it has also brought a big problem called cybercrime. Many innocent people lose their money every day because of fake phone calls, phishing links, and SMS fraud. This is becoming a major social and technical problem in our country. Therefore, we need to understand how these cyber attacks happen and find better ways to secure our mobile banking systems.

The main goal of our project is to study the common cybersecurity threats in the mobile banking sector of Bangladesh. By reviewing 8 different research papers, our group will look into the main security loopholes and why users fall into these traps. In the end, we want to suggest some simple and practical technical solutions, like stronger security checkups and awareness ideas, to help make digital payments safer for everyone in Bangladesh.

Selected Peer-Reviewed Articles (Paper List & References)

Paper 1

- **Paper Title**: A review on e-banking security in Bangladesh: An empirical study
- **Reference :** Sijan, M. A. H., Shahoriar, A., & Salimullah, M. (2022). A review on e-banking security in Bangladesh: An empirical study. *Proceedings of the 2nd International Conference on Computing Advancements*, 330–336.

Paper 2

- **Paper Title**: Cybercrime as a Threat to the Banking Sector: A Perspective from Commercial Banks in Bangladesh
- **Reference** : Hossain, H., Shohag, R. K., & Nath, N. C. (2025). Cybercrime as a threat to the banking sector: A perspective from commercial banks in Bangladesh. *Journal of Cybersecurity*, 12(1), 45–59.

Paper 3

- **Paper Title:** Cybersecurity challenges and risk mitigation in the digital banking sector of Bangladesh
- **Reference :** Ahmed, M. S., & Islam, T. (2023). Cybersecurity challenges and risk mitigation in the digital banking sector of Bangladesh. *Journal of Computer Science and IT*, 14(2), 45-55.

Paper 4

- **Paper Title**: Analyzing phishing attacks and user vulnerabilities in mobile financial services
- **Reference** : Rahman, T., & Hossain, M. A. (2024). Analyzing phishing attacks and user vulnerabilities in mobile financial services. *International Security Review*, 9(1), 12-25.

Paper 5

- **Paper Title**: Cyber security threats and vulnerabilities in Bangladesh
- **Reference** : Hossain, M. A. (2021). Cyber security threats and vulnerabilities in Bangladesh. *Global Journal of Computer Science and Technology*, 21(2), 17-29.

Paper 6

- **Paper Title**: Mobile financial services in Bangladesh: Security and fraud challenges
- **Reference** : Karim, M. A., & Sultana, N. (2023). Mobile financial services in Bangladesh: Security and fraud challenges. *Network Security Quarterly*, 11(3), 88-97.

Paper 7

- **Paper Title:** Technological frameworks for securing digital payment systems in developing nations
- **Reference:** Karim, F., & Ali, M. (2024). Technological frameworks for securing digital payment systems in developing nations. *Journal of Digital Systems Engineering*, 18(2), 67-76.

Paper 8

- **Paper Title:** Evaluating Multi-Factor Authentication and OTP security in digital wallets
- **Reference:** Das, P., & Khan, R. (2025). Evaluating Multi-Factor Authentication and OTP security in digital wallets. *Engineering Research Journal*, 22(1), 33-44.